RAMCO VIDYA MANDIR SR. SEC. SCHOOL, THAMARAIKULAM			
CLASS	**:**	**TIME**	**: 3 Hrs**
EXAM NO	**:**	**MARKS**	**: 80**

Class 11 Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

SECTION A: MCQs (1 mark each) (14 × 1 = 14)

CBSE Class 11 Biology MCQs (Chapter 4 & 5)

1) Which of the following statements regarding Ctenophora is INCORRECT?
a) They exhibit tissue level of organization
b) They have eight external rows of ciliated comb plates
c) Bioluminescence is absent in ctenophores
d) They are exclusively marine organisms

2) Assertion: Ctenophores are commonly known as sea walnuts or comb jellies.
Reason: They possess eight external rows of ciliated comb plates which help in locomotion.
a) Both assertion and reason are true, and reason is the correct explanation of assertion
b) Both assertion and reason are true, but reason is not the correct explanation of assertion
c) Assertion is true but reason is false
d) Both assertion and reason are false

3) Which of the following characteristics is NOT shared by both Porifera and Ctenophora?
a) Diploblastic body plan
b) Marine habitat
c) Intracellular digestion
d) Tissue level of organization

4) In Ctenophora, digestion is:
a) Only extracellular
b) Only intracellular
c) Both extracellular and intracellular
d) Neither extracellular nor intracellular

5) Which feature distinguishes Ctenophora from Porifera?
a) Presence of choanocytes
b) Presence of radial symmetry

c) Presence of bioluminescence
d) Presence of spicules
6) Case study: A marine organism was collected that exhibited radial symmetry, had tissue level organization, possessed eight external rows of ciliated plates for locomotion, and showed bioluminescence. This organism most likely belongs to:
a) Porifera
b) Ctenophora
c) Coelenterata
d) Echinodermata
7) Which of the following is the function of the pulvinus in certain leguminous plants?
a) Protection of the axillary bud
b) Support for the inflorescence
c) Swollen leaf base structure
d) Photosynthesis in the absence of lamina
8) Assertion: In monocotyledonous seeds, the seed coat is generally fused with the fruit wall.
Reason: Monocotyledonous seeds are always endospermic.
a) Both assertion and reason are true, and reason is the correct explanation of assertion
b) Both assertion and reason are true, but reason is not the correct explanation of assertion
c) Assertion is true but reason is false
d) Both assertion and reason are false
9) The region of root where cells undergo rapid elongation and enlargement is:
a) Region of maturation
b) Region of elongation
c) Root cap
d) Meristematic region
10) Which of the following is NOT a characteristic feature of a typical leaf?
a) It is a lateral flattened structure
b) It always develops at the internode
c) It bears a bud in its axil
d) It originates from shoot apical meristem
11) In which type of aestivation do the sepals or petals in a whorl just touch one another at the margin without overlapping?
a) Valvate
b) Twisted
c) Imbricate
d) Vexillary
12) Case study: A plant specimen has the following characteristics - compound leaves with leaflets arranged on both sides of the rachis, reticulate venation, tap root system, and flowers with distinct calyx and corolla. Based on these features, the plant is most likely a:
a) Monocotyledon with pinnately compound leaves
b) Dicotyledon with pinnately compound leaves
c) Monocotyledon with palmately compound leaves
d) Dicotyledon with palmately compound leaves

13) Which of the following statements about floral diagrams is INCORRECT?
a) They provide information about the number of floral parts
b) The position of mother axis is represented by a dot
c) They indicate the fusion of floral parts
d) They represent the color of different floral parts
14) In perigynous flowers:
a) The ovary is situated above the thalamus
b) The ovary is situated below the attachment of other floral parts
c) The margin of thalamus grows upward encircling the ovary
d) The thalamus completely fuses with the ovary wall

SECTION B: Very Short Answer (2 marks each) (10 × 2 = 20)

15) Compare the anatomical differences between dicot and monocot stems in terms of vascular bundle arrangement and explain how this affects their mechanical strength.
16) Analyze how the Casparian strips in the endodermis of roots contribute to the selective absorption of minerals, relating this to the plant's physiological needs.
17) Explain why bulliform cells in grass leaves can be considered an adaptive feature, relating their structure to their function during water stress conditions.
18) Differentiate between racemose and cymose inflorescences in terms of their growth pattern and evolutionary significance.
19) Compare the structural organization of palisade and spongy parenchyma in dicot leaves and explain how these differences relate to their photosynthetic efficiency.
20) Evaluate how the arrangement of xylem and phloem in a conjoint vascular bundle provides advantages for the transport of materials in flowering plants.
21) Explain how the aestivation patterns in flowers (like valvate, twisted, and imbricate) reflect evolutionary adaptations related to protection and pollination.
22) Analyze how the internal structure of a dorsiventral leaf demonstrates structural adaptation to maximize photosynthesis while minimizing water loss.
23) Compare the anatomical features of monocot and dicot roots in relation to their ability to undergo secondary growth, with reference to the presence or absence of specific tissues.
24) Evaluate how the arrangement and structure of stomatal apparatus in the leaf epidermis facilitates gas exchange while preventing excessive water loss.

SECTION C: Short Answer Type I (3 marks each) (7 × 3 = 21)

25) Explain how the arrangement of vascular bundles in dicot stems contributes to their ability to undergo secondary growth, and contrast this with the arrangement in monocot stems. Include specific anatomical features in your explanation.
26) Compare and contrast the adaptations in leaf anatomy of monocot and dicot plants, focusing on how these adaptations relate to their respective environments and physiological processes.
27) Analyze the structural features of the endodermis in plant roots and explain how these features functionally contribute to the selective absorption and transport of water and minerals.
28) Describe how the anatomical differences between monocot and dicot roots reflect their

evolutionary adaptations, with specific reference to the arrangement of vascular tissues and potential for secondary growth.

29) Evaluate the significance of bulliform cells in grass leaves, explaining their structure, function, and ecological importance in water conservation during environmental stress conditions.

30) Explain the functional relationship between the tissue systems in a plant stem, focusing on how their arrangement provides both mechanical support and efficient transport of materials.

31) Analyze how the internal structure of dorsiventral leaves reflects adaptation to maximize photosynthetic efficiency while minimizing water loss, with reference to specific tissues and their arrangement.

SECTION D: Case-based/Competency (4 marks each) (2 × 4 = 8)

32) Based on the phylum characteristics and experimental data, explain the possible relationship between oxygen levels and bioluminescence in ctenophores. (2 marks)

33) Considering the tissue-level organization of ctenophores and the data from Group C, analyze how their cellular structure might enable them to respond to mechanical stimulation with increased bioluminescence. (2 marks)

Hard Case-Based Question 2 (4 marks)

Case Study: Comparative Leaf Anatomy and Environmental Adaptation

Botanist Dr. Sharma collected leaf samples from two different plant species growing in the same botanical garden - Plant X (a dicot) and Plant Y (a monocot). She prepared microscopic slides of their transverse sections and observed the following features:

Plant X:

- Distinct upper and lower epidermis with more stomata on lower surface
- Clearly differentiated palisade and spongy mesophyll
- Reticulate venation with vascular bundles of varying sizes
- Thick cuticle on upper epidermis
- Average stomatal density: 120 per mm^2

Plant Y:

- Similar upper and lower epidermis with nearly equal stomatal distribution
- Homogeneous mesophyll (not differentiated into palisade and spongy layers)
- Parallel venation with vascular bundles of similar sizes
- Presence of bulliform cells on the upper epidermis
- Average stomatal density: 85 per mm^2

Dr. Sharma then subjected both plants to drought conditions for two weeks and measured their water loss and photosynthetic efficiency:

Parameter	Plant X (Before)	Plant X (After)	Plant Y (Before)	Plant Y (After)
Water loss rate (mL/day)	25	15	30	12
Photosynthetic efficiency (%)	90	75	85	60
Leaf curling	None	Slight	None	Significant

Questions:

SECTION E: Long Answer (5 marks each) (3 × 5 = 15)

LONG ANSWER DESCRIPTIVE QUESTIONS (5 marks each)
Question 1
Discuss the unique characteristics of phylum Ctenophora, including their body organization, locomotion mechanism, and bioluminescence. Explain how these features distinguish them from other marine organisms.
OR
Describe the various levels of organization observed in the animal kingdom, from cellular to organ-system level. Explain how this organizational complexity relates to evolutionary advancement with suitable examples from different phyla.
Question 2
Explain the morphology and anatomy of a dicotyledonous leaf with special reference to its internal structure. Discuss how the arrangement of tissues in dorsiventral leaves facilitates photosynthesis and gas exchange.
OR
Compare and contrast the anatomical features of monocotyledonous and dicotyledonous stems with the help of labeled diagrams. Explain how these structural differences reflect their evolutionary adaptations.
Question 3
Describe the digestive and respiratory systems of frog in detail. Explain how the frog's respiratory mechanism allows it to survive both in aquatic and terrestrial environments.
OR
Elaborate on the reproductive system of male and female frogs. Discuss the process of fertilization, development, and metamorphosis in frogs, highlighting their ecological significance in the ecosystem.

QUESTION PAPER GENERATION SUMMARY

Class: 11

Subject: Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

Section Details:

Section A (MCQs): 14 questions | 1 mark each | Total: 14 marks

Chapters: 4, 5 | Context: 27643 chars

Section B (Very Short Answer): 10 questions | 2 mark each | Total: 20 marks

Chapters: 5, 6 | Context: 26654 chars

Section C (Short Answer Type I): 7 questions | 3 mark each | Total: 21 marks

Chapters: 6, 7 | Context: 27717 chars

Section D (Case-based/Competency): 2 questions | 4 mark each | Total: 8 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars

Section E (Long Answer): 3 questions | 5 mark each | Total: 15 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars